Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/29/2020
PI Name:	Gervais, Luc Ph.D.		
Project Title:	Miniaturized Diagnostics Platform for Health Assessment in Deep Space		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	luc.gervais@1dropdx.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	857-777-9157
Organization Name:	1Drop Diagnostics		
PI Address 1:	21 Drydock Ave Fl. 6		
PI Address 2:			
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02210	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Industry
Start Date:	09/01/2018	**End Date:**	08/31/2019
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-IND0001		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Industry Project 1Drop is a next-generation point-of-care diagnostics device that performs sensitive, specific, and rapid detection of multiple blood based biomarkers. The goal of this project will be to further miniaturize the device and extend the capabilities of the platform with an expanded panel of biomarkers. Easy-to-use and powerful blood biomarker detection will enable NASA to monitor the health status of astronauts. This project lead by Dr. Gervais is developing a small, hand-held blood analysis platform to assess liver and kidney biomarkers in blood. The platform is easy to use, and its cartridges are room temperature stable. For an exploration-class manned space flight mission with limited operational resources, Dr. Gervais' project meets the goals of optimizing operational resources while providing a capability that performs required blood analysis functionality.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1DROP is based on novel microfluidics, biochemistry, photonics, and machine learning that enable multiplexing (simultaneous detection) of up to 64 biomarkers in a short time with high (picomolar) sensitivity, accuracy, precision, and portability. Only a tiny drop of capillary blood is required, results are ready in 15 minutes, and are accessible to people and healthcare practitioners via the 1DROP application. 1DROP will enable precision medicine through non-invasive diagnostic tests, and provide health information that is accessible to everyone everywhere, even in space. The game-changing 1DROP device will transform the diagnostics and management of disease by: 1) Diagnosing disease at an early stage, 2) Empowering people to improve their health and prevent disease, 3) Optimizing treatment, 4) Improving clinical workflow, 5) Improving people's quality of life, and 6) Lowering the cost of people's care. 1DROP has developed a ground-breaking device that can detect multiple biomarkers after the addition of one drop of sample, without the need for peripheral equipment for fluid control and processing. In this project, we optimized and scaled up production of the biochips and photonics reader, and developed new tests of kidney and liver function. Ultimately, this project has led to the development of a novel, highly sensitive, highly accurate, precise, and rapid diagnostic assay; and importantly demonstrated the potential of the platform that is adaptable to a much broader scope of clinical applications: sepsis, liver/kidney/thyroid function, dementia, iron deficiency, fertility, diabetes, autoimmune disease, oncology, infectious disease.
Task Progress:	1DROP is a next-generation point-of-need diagnostics device that performs sensitive, specific, accurate, precise, and rapid detection of multiple blood based biomarkers. The goal of this project was to further miniaturize the device and extend the capabilities of the platform with an expanded panel of biomarkers of high priority analytes of liver and kidney function: creatinine, alkaline phosphate (ALP), and aspartate aminotransferase (AST). Easy-to-use and powerful blood biomarker detection will enable NASA to monitor the health status of astronauts. This project was to develop a small, hand-held blood analysis platform to assess liver and kidney biomarkers in blood. The platform is easy to use and its cartridges are stable at room temperature. For an exploration-class manned spaceflight mission with limited operational resources, the platform meets the goals of optimizing operational resources while providing the capability to perform the required blood analysis functionality.
Bibliography Type:	Description: (Last Updated:)